Group Assignment

Machine Learning for Business (QBUS6810)

1 Overview

Your team acts as an analytics consultancy advising the residential valuation team at the Cook County Assessor's Office. Your task is to develop and evaluate a regression model that estimates a home's selling price from its characteristics and location.

Cook County is a county in the US state of Illinois. It includes Chicago and surrounding suburbs. The Assessor's Office is a local government agency. It uses estimates of what properties would sell for to help assess them for tax purposes. A home does not need to be on the market for the office to assess it.[1]

You are given historical transactions with known sale prices. Use these records to train and test your model, with `sale_price` in US dollars as the outcome. The proposed use is to provide an initial price estimate for a home without a recent sale, for a property analyst to review. Recommend whether your results support this use, considering prediction errors and the limits of learning only from homes that sold.

2 Data

The supplied public records contain completed sales and property characteristics, including building size, age, location, nearby facilities and neighbourhood information. They do not contain property listings or asking prices. Homes without a recorded sale in this sample are not included; their sale prices are unobserved.

Use the two datasets in the `data` folder:

- `sales.csv`: 109,307 sales from 2023–2025, with property and transaction identifiers, sale date and sale price.
- `properties_2022.csv`: 98,899 properties, with a property identifier and 97 property and neighbourhood features.

Join on `property_id`. Each sale should match one property record. A property may have several sales. Read `QBUS6810_data_dictionary.pdf` and `variable_dictionary.csv` for field definitions. All required data are supplied as CSV files.

You may supplement the supplied data with external datasets. For each additional dataset, give its provider, dataset name, source URL and access date in your report. Explain its relevance and how you combine it with the supplied data.

The target is `sale_price`. Property features come from the 2022 archive, before the sales. The `sale_date` field provides calendar information for analysing prices over time. Do not use `sale_price`, `property_id` or `transaction_id` as input features. Keep the identifiers for joining data and checking repeated sales.

[1] Cook County Assessor's Office: Home Value Report and explanation of residential valuation models.

3 Deliverables

Submit to Canvas:

1. **Report (PDF):** your analysis, reasoning and recommendations, following the structure below.

2. **Code:** a Jupyter notebook or Python script, in a maximum of two files, that reproduces the reported results.

3. **Additional data (if used):** upload the external data files used in your analysis so that your results can be reproduced.

4 Report structure

Use the following five sections. Answer the numbered questions; related answers may share a paragraph, table or figure. Communication and code quality are assessed across the submission. Include AI reflection in a short appendix.

Section 1: Business objective

1a. Explain the client's business problem, who will use the predictions, and the practical value the project could provide.

Section 2: Data understanding and representation

2a. What does each row represent? Verify the join, check for duplicate records and repeated sales of the same property, and justify how you handle them.

2b. Which missing or unusual predictor values matter, and how will you handle them?

2c. What do summaries and plots show about prices over time and across locations? Identify higher- and lower-price periods and discuss possible explanations.

2d. How will you represent the data? Create and justify at least two new features.

2e. What could cause leakage or biased conclusions? How does using only sold properties limit claims about homes that have not sold?

Section 3: Model development

3a. Establish a simple price-prediction baseline and explain its purpose.

3b. Compare multiple regression approaches taught in the course. Explain your choices and what you learned. You may also investigate separate models for different locations.

3c. How will you split the data and compare models? Justify your validation strategy, considering sale dates and repeated properties.

3d. Show how experiments informed your decisions. Include a comparison of feature sets.

3e. How did you tune the models and control complexity?

3f. Which final model and prediction procedure did you select, and why?

Section 4: Final evaluation

4a. How do your price predictions perform on the test set compared with the baseline? Justify your regression metrics and interpret a residual plot.

4b. Compare performance across time periods and at least three townships (local geographical areas in Cook County). How might price levels and the mix of homes sold explain differences? Support your explanations with evidence from the data.

4c. Choose and justify weights for overvaluation and undervaluation. Compare weighted absolute error for your model and the baseline under your main scenario and two alternatives. Keep predictions fixed across scenarios. Does your conclusion change?

4d. Do the results justify a pilot of your model to support property valuers? Explain your recommendation using the errors, practical value and limitations.

For Question 4c, the following error weights are options. You may instead propose weights supported by relevant research. These weights express priorities, not actual financial costs.

Scenario	Overvaluation	Undervaluation
Equal priority	1	1
Greater concern about overvaluation	2	1
Greater concern about undervaluation	1	2

For weighted absolute error, multiply each absolute price error by the weight for its direction, then average over sales. Choose your main scenario before model selection and evaluate two alternatives in Question 4c. Cite any research used.

Section 5: Deployment and operations

5a. How would the office generate, review and use estimates for homes without recent sales? Identify the information and checks needed before an estimate is adopted.

5b. What would you monitor after introducing the model? Explain what findings would trigger investigation, retraining or increased human review.

5c. Identify two important risks or constraints when using your model in practice. Explain how the organisation should manage them.

5d. For the case below, should the valuer simply subtract the repair quote from the model estimate? Explain your decision and the further evidence needed in at most 150 words.

Hypothetical case for 5d. Your model estimates a home's value at US$400,000 using archived property information. The owner supplies a recent inspection showing major damage and a repair quote of US$30,000. The damage is not recorded in the model inputs.

5 AI use and reflection

Include four short cases in an appendix, using the following labels:

7a. Data or features.

7b. Validation or leakage.

7c. Modelling or debugging.

7d. Business interpretation.

For each case, answer all four points:

1. What did you initially think? Give the AI tool, date, prompt or relevant excerpt, and a brief summary of its advice.

2. How did you check the advice, and what did the check show? Refer to your code, results or a reliable source. At least two cases must use your data or code.

3. What did you accept, change or reject, and why? Explain the effect on your work.

4. What did you learn, and what remains uncertain?

Use at most 180 words per case and two appendix pages in total. Do not paste full conversations. Add a short declaration of other AI assistance.

6 Formatting and submission

- Page limit: 18 pages, excluding appendices. Keep explanations relevant and concise.
- Use the five report headings above. Label and discuss all figures and tables.
- Put additional experiment logs and supporting figures in an appendix. Keep the evidence for your main conclusions in the report.
- Acknowledge data and other sources. Include a short group contribution statement.
- Deadline: 06 Nov, 23:59 (Week 13). Follow Canvas for group size and submission details.

7 Assessment and reproducibility

The accompanying rubric gives the marks and marking criteria. Support your decisions with evidence from this dataset.

Your code must run in a clear order using relative file paths, fixed random seeds where relevant and no hidden manual steps. List required packages and brief running instructions at the start of the notebook or script. Remove unused code and irrelevant output.